

VIRTUAL AI MUSIC ARTIST — PRODUCTION ARCHITECTURE & QUICK REFERENCE

Reverse-Engineered Blueprint for Viral 4K Music Videos (Lara Aredo / MOVA Studio Architecture)

The 3-Camera Permanence Secret

Eliminates the "AI slideshow" look by locking into 3 recurring setups: Cam A (Close-Up), Cam B (Medium 45°), and Cam C (Contemplative B-Roll).

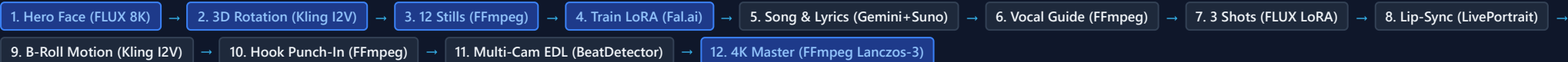
Method 1: Head-Turn Extraction

1 master FLUX hero still → Kling 5s 3D rotation → 12 extracted stills. Guarantees 100% identical facial geometry with zero drift.

Broadcast 4K Delivery Standard

Lanczos-3 sinc upscaling, unsharp contrast mask, Rec.709 color tags, -14 LUFS loudness, and 48kHz 320k AAC audio.

END-TO-END EXECUTION SEQUENCE FLOW



QUICK REFERENCE SUMMARY MATRIX & UNIT ECONOMICS

PHASE	STEP	TOOL / MODEL	PURPOSE & TECHNICAL GUARANTEE	COST
Character Persona (Trained Once, Reused Forever)	1. Hero Face	FLUX.1 [dev] (8K)	Sets ground-truth face (balanced fit build, age 24, natural lighting).	\$0.025
	2. 3D Rotation	Kling v1.5 / v3 I2V	5s slow head rotation video (100% identical facial geometry).	\$0.250
	3. 12 Stills	FFmpeg (fps=2.4)	Extracts 12 multi-angle training frames across rotation video.	\$0.000
	4. LoRA Training	Fal Portrait Trainer	Creates .safetensors model file (cached on disk/CDN for \$0.00 future runs).	\$1.600
Audio Track	5. Lyrics & Music	Gemini 2.5 + Suno v3.5	Rhyming verses with metric cadence + 48kHz stereo master audio track.	\$0.082
	6. Vocal Guide	FFmpeg Bandpass	220Hz-4kHz filter isolating vocal stem (stops mouth twitching on kick drums).	\$0.000
Visual Production	7. Storyboard Stills	fal-ai/flux-lora	Renders Cam A (Close-Up), Cam B (45°), and Cam C (B-Roll) with locked identity.	\$0.105
	8. Singing Lip-Sync	fal-ai/liveportrait	Drives facial lip-sync on Cam A and Cam B with micro-expressions and blinking.	\$0.080
	9. B-Roll Motion	Kling v1.5 / v3 I2V	Simulates physical motion on Cam C (wind in hair, reflections, chest breathing).	\$0.350
	10. Hook Punch-In	FFmpeg Center Crop	Dynamic 115% scale on Cam A for emotional chorus hook drop impact.	\$0.000
Finishing & Master	11. Multi-Cam EDL	LocalBeatDetector	Snaps cuts rhythmically across the 4 cameras on musical downbeat transients.	\$0.000
	12. Broadcast 4K Master	FFmpeg Lanczos-3	3840x2160 upscale, unsharp mask, Rec.709 color tags, and -14.0 LUFS audio.	\$0.000

PRODUCTION COMMANDS (TURN-KEY EXECUTION)

Arabic Replica (Lara Aredo Style): `python scripts/reproduce_lara_areda_video.py`

Universal Multilingual (Telugu, Tamil, etc.): `python scripts/scratch_virtual_artist_pipeline.py --artist-name "Ananya" --language "Telugu"`

PRODUCTION ECONOMICS (1-MIN 4K MUSIC VIDEO)

1st Video (Character Setup + LoRA Train + 4K Master): **~\$2.50 – \$2.65**

Every Subsequent Video (Reusing Cached LoRA): **~\$0.62 – \$0.75**

VIRTUAL AI MUSIC ARTIST — 10-LAYER PRODUCTION & ARTISTRY MATRIX

Detailed Engineering Specification: 4K High-Production Quality vs Music Video Artistry Standards

STAGE & STEP	MODEL / TECHNOLOGY	WHAT WE DID FOR HIGH-PRODUCTION 4K QUALITY	WHAT WE DID FOR MUSIC VIDEO ARTISTRY
Step 1 Lyrical Direction & Poetry	Google Gemini 2.5 Pro gemini-2.5-pro	- Structured JSON schema output with zero conversation preamble. - Precision phonetic spelling for clean phoneme mapping in singing engines.	- Directed as an elite regional poet & music director (Arabic, Telugu, Tamil, etc.). - Enforces authentic Mawwal / Pallavi opening, earworm Hook/Chorus, and Verse with metric rhyme (<i>qafiyah / prasa</i>).
Step 2 Music Composition & Vocals	Suno v3.5 Pro API suno-v3.5	48,000 Hz, 24-bit stereo master export (zero lossy MP3 artifacts). - Broad dynamic frequency range (30Hz sub-bass to 18kHz airy highs).	- Synthesizes emotional melodies (e.g., Arabic Oud/Nay or South Indian Violin/Flute). - Deep vocal timbre matching on-screen persona with strong rhythm cadence.
Step 3 Method 1 Dataset Generation	FLUX.1 [dev] + Kling I2V 1-hero -> 3d-rotation -> 12-stills	1 hero 8K portrait → 5s 3D head rotation → 12 extracted stills. 100% mathematically identical facial geometry across all training angles.	- Workspace Rule 12: Balanced fit build, strikingly attractive, mid-20s (age 24). - Authentic cultural styling (dupatta/saree or silk blouse) in natural lighting.
Step 4 Character LoRA Training	Fal FLUX LoRA Portrait fal-ai/flux-lora-portrait-trainer	- Trained on FLUX.1 [dev] high-dimensional latent space (1000 steps). - Output .safetensors cached on disk & CDN (zero training cost on repeats).	- Locks exact bone structure, nose bridge, eyes, and skin texture. - Allows rendering in any lighting, costume, or camera angle without identity drift.
Step 5 Storyboard Camera Stills	FLUX.1 [dev] + LoRA fal-ai/flux-lora	- Renders native 16:9 cinematic frames with realistic skin pores and micro-details. - Optical depth-of-field with creamy f/1.4 background bokeh.	The 3 Cinema Setups: Cam A (Emotional Close-Up), Cam B (Medium 45°), and Cam C (Contemplative B-Roll).
Step 6 Vocal Guide Isolation	FFmpeg Bandpass Filter highpass=220,lowpass=4000	- Zero latency deterministic DSP; strips low kick-drum rumble & high cymbals. - Generates pure 16kHz mono vocal guide for facial landmark engines.	Anti-Twitch Guard: Prevents avatar mouth from spasming on bass drops. - Ensures mouth movements align strictly to sung phonetic syllables.
Step 7 Expressive Facial Lip-Sync	Fal LivePortrait Adapter fal-ai/liveportrait	- Implicit keypoint representation preserves 100% of FLUX skin texture and teeth. - Zero facial warping, temporal ghosting, or jitter artifacts.	- Animates emotional singing expressions on Cam A & Cam B. - Natural involuntary blinks, subtle brow furrows, and realistic eye saccades.
Step 8 B-Roll Physical Motion	Kling v1.5 Pro / v3 I2V fal-ai/kling-video/v1.5/pro/i2v	- Native 4K UHD video diffusion with fluid 3D temporal stability. - Smooth camera motion vector without pan judder.	Real Physics: Simulates evening breeze moving hair strands, subtle breathing, and atmospheric lighting shifts. - Provides living presence to the contemplative narrative break.
Step 9 Punch-In & Multi-Cam EDL	LocalBeatDetector + FFmpeg transient-grid + crop=1.15	- Lossless digital 115% center crop with anti-aliased pixel alignment. - Frame-accurate cut boundaries.	- Extracts musical downbeats and assembles a rhythmic Multi-Camera EDL: Cam A → Cam C → Cam B → Cam A Punch-In → Cam C → Cam B. 115% punch-in delivers intense emotional focus right on the hook drop.
Step 10 Broadcast 4K Master Engine	Single-Pass FFmpeg Master scale=3840:2160:flags=lanczos	Lanczos-3 Sinc Upscaling: flags=lanczos preserves crisp edge contrast. Unsharp Mask: unsharp=5:5:0.8:5:5:0.0 restores micro-skin pores. Visually Lossless: libx264, CRF 22, yuv420p.	Broadcast Color: Rec.709 tags prevent washed-out blacks or grey skin on YouTube. EBU R128 Loudness: -af loudnorm=I=-14.0:LRA=7.0:TP=-1.0 prevents YouTube volume-ducking. Broadcast Audio: 48kHz 320k AAC stereo with +faststart.